

SUMMARY

AI Research Engineer with 2+ years of experience developing scalable software systems and applied machine learning solutions. Skilled in LLM fine-tuning, NLP, model evaluation, data pipelines, and cloud-native application development. Passionate about leveraging AI to solve real-world problems through robust engineering, reproducible experimentation, and data-driven decision-making.

SKILLS

Languages: JavaScript, TypeScript, Python, Dart, Java

Frameworks: Flutter, React.js, Next.js

Backend: Node.js, Express.js, FastAPI

Databases: PostgreSQL, MongoDB, Redis, DB2

Cloud & DevOps: AWS, Firebase, GitHub Actions, Cloudflare

AI/ML: Deep Learning, Neural Networks, LLM fine-tuning, RAG systems, Data Preprocessing

Libraries/Tools: PyTorch (basic), NumPy, Pandas

PUBLICATIONS

A Multi-Curricula Dataset and Benchmark for Elementary-level Math Word Problem Generation

Co-author — Submitted to Language Resources and Evaluation (Springer), April 2026

EXPERIENCE

Project Associate I — IIT Ropar

Oct 2025 – Present

- Fine-tuned large language models and experimented with neural network-based architectures for improving task-specific performance.
- Built automated evaluation pipelines for benchmarking AI models across NLP tasks using structured datasets.
- Designed data preprocessing and experimental workflows including dataset curation, cleaning, and inference testing.
- Worked with Python-based ML workflows, focusing on model iteration, performance analysis, and reproducibility.

Mobile App Developer — HBCH&RC, New Chandigarh

Mar 2024 – Oct 2025

- Built and maintained a production Flutter mobile application enabling digital hospital services for patients.
- Designed and implemented REST APIs using Node.js and TypeScript integrated with hospital information systems.
- Deployed and managed backend services on IIS servers ensuring reliable system uptime.

PROJECTS

Afrolingual — Multilingual Translation System *NLLB, FastAPI, Speech Processing*

- Developing a multilingual translation system to convert English and French into African local languages using Meta's NLLB-600 model.
- Designing API-based inference system using FastAPI for scalable deployment and real-time translation.

Cow Neck Health Monitoring System (Ongoing) *Kotlin, FastAPI, React.js, PostgreSQL, BLE*

- Developing a livestock health monitoring platform using BLE-enabled accelerometer devices attached to cattle.
- Built a mobile application to collect sensor telemetry and synchronize data to cloud services through FastAPI APIs.
- Developed a web dashboard for visualizing activity patterns and supporting ML-based health inference workflows.

AnnoForge — Creator (Copyright Under Process) *FastAPI, Node.js, Tauri, SQLite*

- Developed a desktop annotation platform supporting multi-project and multi-user dataset management.
- Integrated AI-assisted semi-automatic labeling workflows through model inference pipelines.
- Enabled efficient dataset preparation for computer vision tasks including object detection and segmentation.

TMC Punjab Saarthi — Full-Stack Developer

Flutter, Node.js, Express, DB2, Firebase

- Built a hospital app for appointments and patient records.
- Developed secure backend APIs integrated with hospital systems.

EDUCATION

Master of Computer Applications (MCA)

2022 – 2024

Chandigarh University (*CGPA: 8.03*)

Bachelor of Computer Applications (BCA)

2018 – 2021

Dr. B. R. A. Bihar University (*Score: 80%*)

ACHIEVEMENTS

- **1st Place** — Problem Solving Competition, GDSC Chandigarh University (2023)
- **2nd Place** — Project Expo for Waste2Wealth (2023)
- **3rd Place** — Talent Hunt (Web Development), Uniford Foundation (2023)